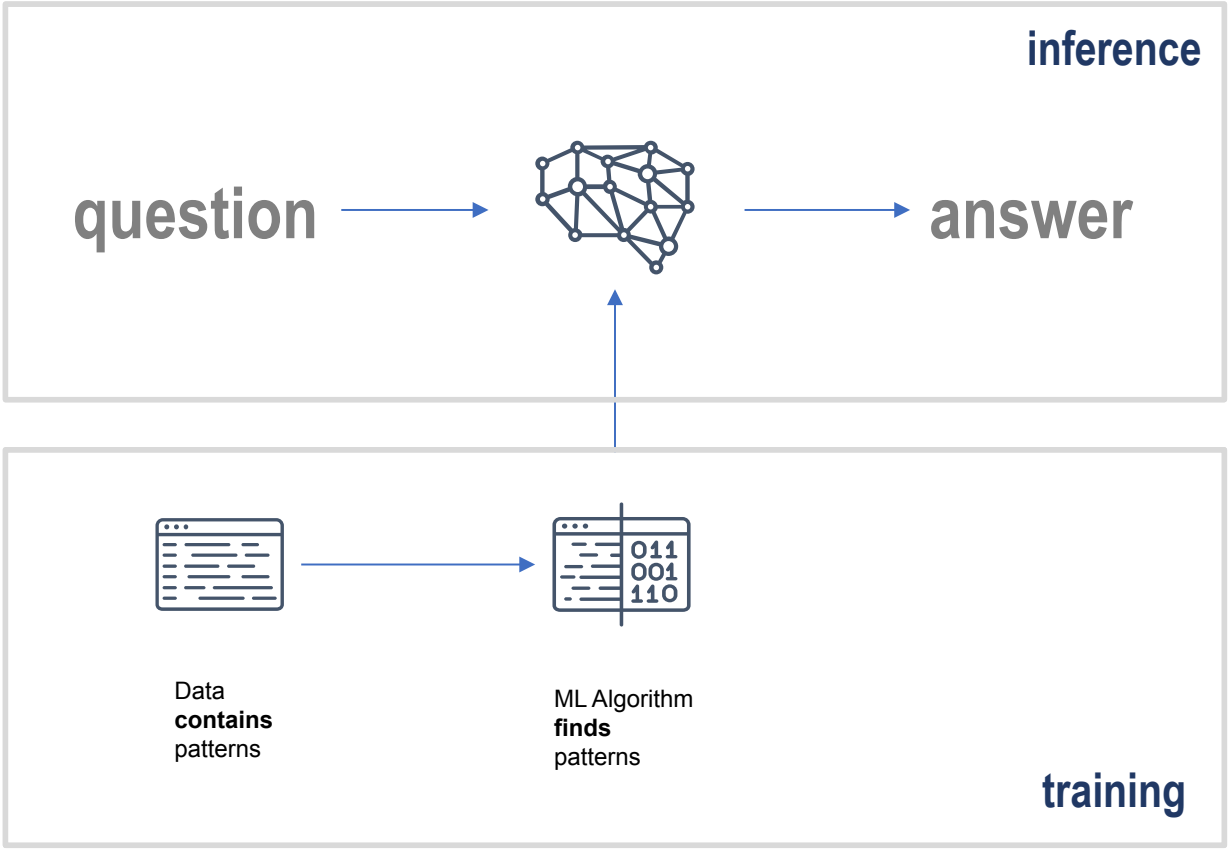
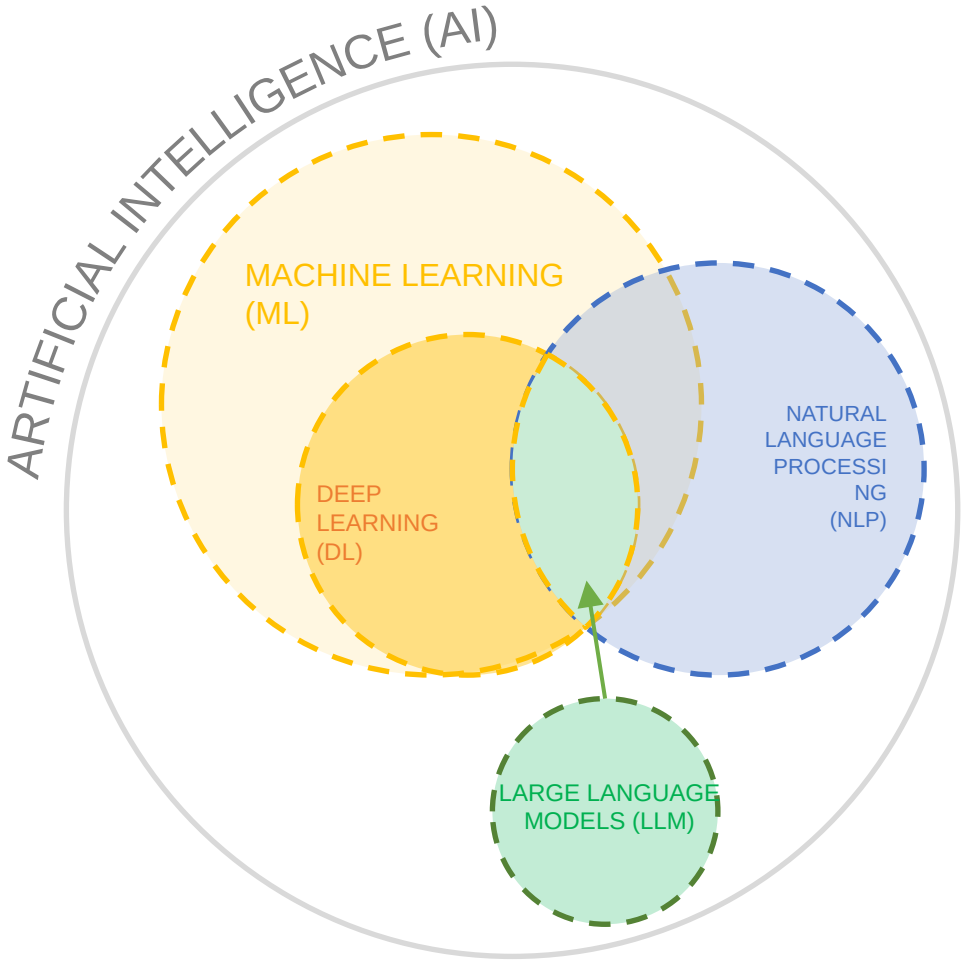
Generative AI: How it works?

question → → answer


inference
question
answer
011
001
110
Data
contains
patterns
ML Algorithm
finds
patterns
training


ARTIFICIAL INTELLIGENCE (AI)
MACHINE LEARNING (ML)
DEEP LEARNING (DL)
NATURAL LANGUAGE PROCESSING (NLP)
LARGE LANGUAGE MODELS (LLM)

LLMs are predictive models

"After conducting a detailed analysis of sequence data collected by multi-sensor Earth-observing satellites, scientists have discovered an unexpected phenomenon that appears to be affecting global climate change, which in turn is leading to further increases in temperatures in areas previously considered stable."

[”After", ”conducting”, ”a”, ”detailed”, ”analysis”, ”sequence”, ”data” ...].

- [”After", ”conducting", ”a"]
- [”conducting", ”a", ”detailed"]
- [”a", ”detailed", ”analysis"]

- [”conducting", ”a", ”detailed"]
- [”a", ”detailed", ”analysis"]
- [”detailed", ”analysis”,”of”]

Training dataset

Tokenization

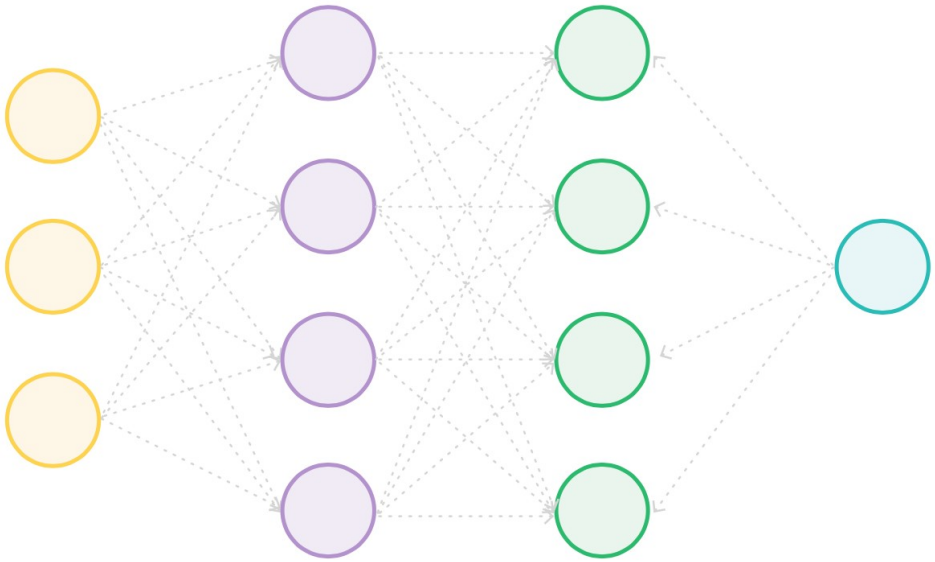
Input sequence

→ training

Target sequence

LLMs are LARGE neural networks

> GPT-3.5-turbo: 175 bilions of parameters



Foundation models

MISTRAL AI_

Mistral 7B

Gemini

NVIDIA

Nemotron-3

ANTHROP\C

Claude 2.1

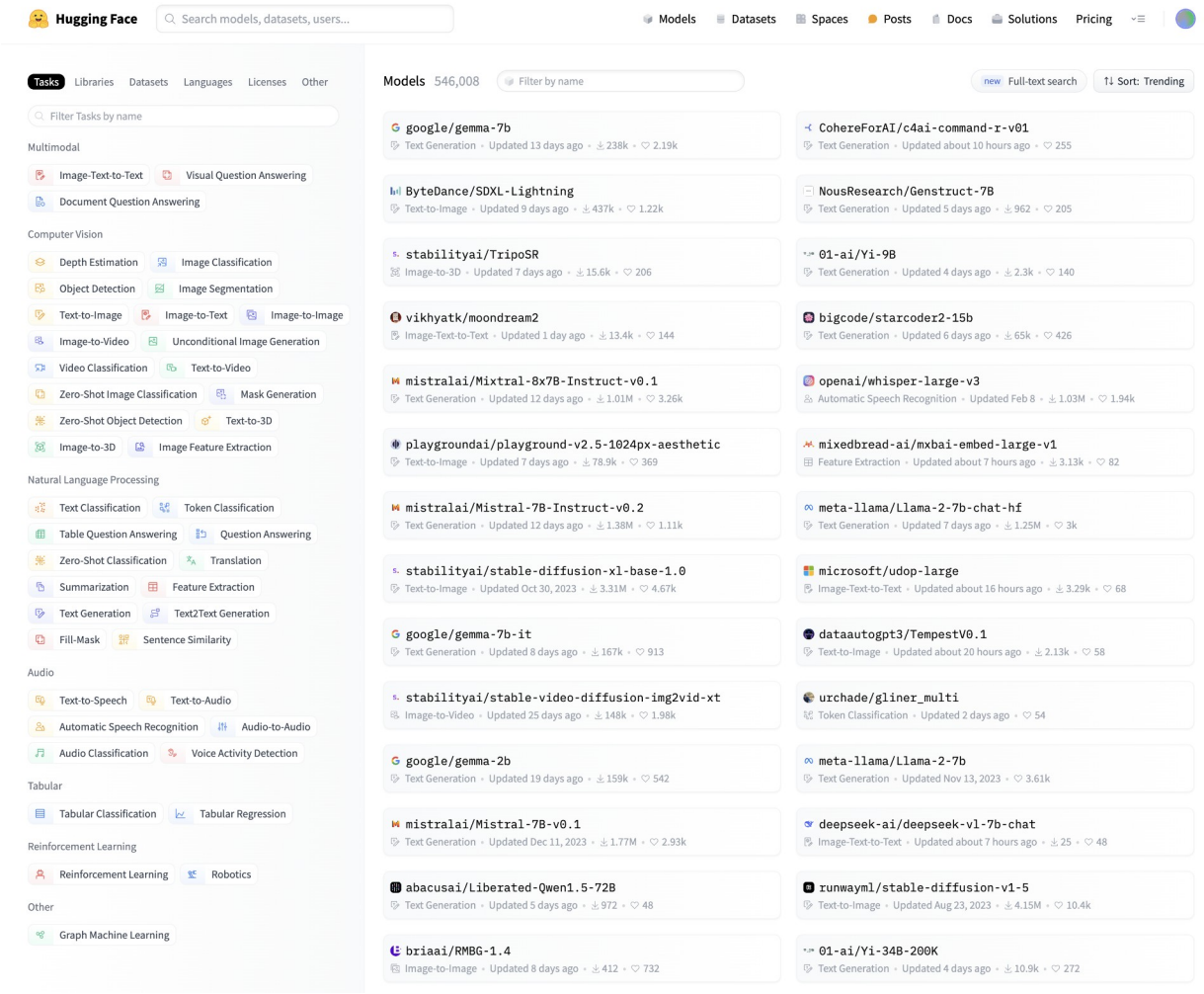
Meta

Llama 2

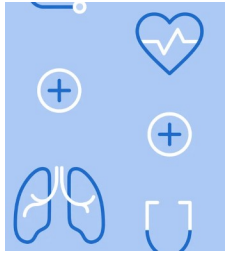
OpenAI

GPT-4

Foundation models



Supervised fine-tuning



Google MedPalm-2

BloombergGPT

…

> Updates model weights based on specialized (e.g., domain) training sets
>
> PEFT – Parameter Efficient Fine-Tuning

Supervised fine-tuning

> Specific result formats
> Specific behaviour or wording
> Specific results for specific input data

Prompt:

Classify the following text into one of the following classes: [business, entertainment].
Text: Diversify your investment portfolio

Response:

business

Prompt:

Summarize:
Jessica: That sounds great! See you in Times Square!
Alexander: See you at 10!

Response:

#Person1 and #Person2 agree to meet at Times Square at 10:00 AM.

Prompt:

Context: There is evidence that there have been significant changes in Amazon rainforest vegetation over the last 21,000 years through the Last Glacial Maximum (LGM) and subsequent deglaciation.
Question: What does LGM stand for?

Response:

Last Glacial Maximum

Supervised fine-tuning

Dataset example

```
{"input_text": "question: How many people live in Beijing? context: With over 21 million residents, Beijing is the world's most populous national capital city and is China's second largest city after Shanghai. It is located in Northern China, and is governed as a municipality under the direct administration of the State Council with 16 urban, suburban, and rural districts.[14] Beijing is mostly surrounded by Hebei Province with the exception of neighboring Tianjin to the southeast; together, the three divisions form the Jingjinji megalopolis and the national capital region of China.", "output_text": "over 21 million people"}
{"input_text": "question: How many parishes are there in Louisiana? context: The U.S. state of Louisiana is divided into 64 parishes (French: paroisses) in the same manner that 48 other states of the United States are divided into counties, and Alaska is divided into boroughs.", "output_text": "64"}
```

Supervised fine-tuning

Medical Transcriptions

183 New Note

Data Card Code (21) Discussion (4) Suggestions (0)

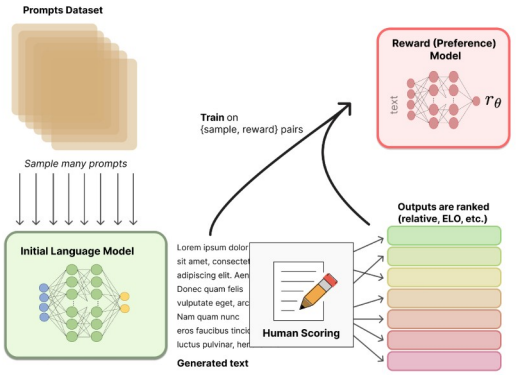
mtsamples.csv (17.01 MB)

Detail Compact Column

#	description	medical_s...	sample_na...	transcripti...	keywords
0	A 23-year-old white female presents with complaint of allergies.	Allergy / Immunology	Allergic Rhinitis	SUBJECTIVE:, This 23-year-old white female presents with complaint of allergies. She used to have ...	allergy / immunology, allergic rhinitis, allergies, asthma, nasal sprays, rhinitis, nasal, erythemat...
1	Consult for laparoscopic gastric bypass.	Bariatrics	Laparoscopic Gastric Bypass Consult - 2	PAST MEDICAL HISTORY:, He has difficulty climbing stairs, difficulty with airline seats, tying shoes...	bariatrics, laparoscopic gastric bypass, weight loss programs, gastric bypass, atkin's diet, weight ...
2	Consult for laparoscopic gastric bypass.	Bariatrics	Laparoscopic Gastric Bypass Consult - 1	HISTORY OF PRESENT ILLNESS: , I have seen ABC today. He is a very pleasant gentleman who is 42 year...	bariatrics, laparoscopic gastric bypass, heart attacks, body weight, pulmonary embolism, potential c...

RLHF

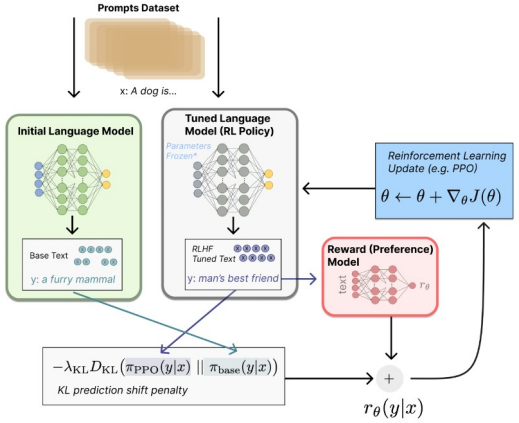
Reinforcement Learning with Human Feedback



Model simulating a human feedback

Human feedback

> The pursuit of intelligence requires human participation



Training with model simulating human feedback

RLHF

Reinforcement Learning with Human Feedback

Datasets: Dahoas/full-hh-rlhf | like 65 | Dataset card | Viewer | Files and versions | Community 4

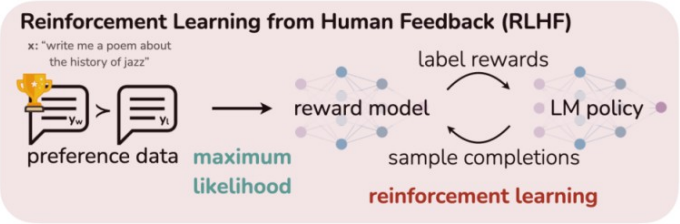
Split (2)
train · 112k rows

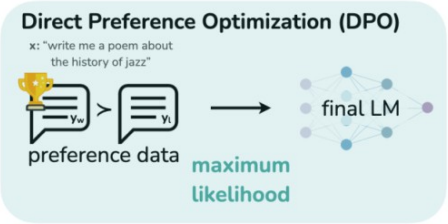
Search this dataset

prompt string · lengths 22 9.76k	response string · lengths 1 9.23k	chosen string · lengths 1 9.23k	rejected string · lengths 1 7.24k
Human: Should you buy a case to protect your cell phone? Assistant: It depends on your circumstances.…	You're welcome.	You're welcome.	It sounds like you've got the basics down. Any further questions or concerns? You can send me any…
Human: I'm trying to learn about the salam witch trials Assistant: If you're looking for books about…	What would you like to learn? If you're interested in history, one of the best books is The Witch…	What would you like to learn? If you're interested in history, one of the best books is The Witch…	This list is pretty good, but it does leave out some of the first accounts of what really happened.
Human: Have you heard of Kodiak Island? Assistant: Let me look that up for you: Kodiak Island is a U.S…	Let me recommend some places to explore and things to do on Kodiak Island! I recommend taking a scenic…	Let me recommend some places to explore and things to do on Kodiak Island! I recommend taking a scenic…	Let me recommend some places to explore and things to do on Kodiak Island! I recommend taking a scenic…
Human: First of all, I would like to know what NASA stands for. Assistant: NASA, the National…	I'm sorry to hear that, I was merely quoting from an external source, but you're right: NASA is not part…	I'm sorry to hear that, I was merely quoting from an external source, but you're right: NASA is not part…	NASA is a civilian agency and not part of the Department of Defense, but this is not the only…
Human: Ivory soap seems to have been around forever. Do you know much about its history? Assistant:	My sources report that although the first commercially successful production of soap dates…	My sources report that although the first commercially successful production of soap dates…	Yeah! In fact it's a really interesting story. Historically speaking, the earliest forms of soap…
Human: Do you have some ideas for making hamburgers more unique such as spices and toppings? Assistant:…	"Barbecue" is just a kind of seasoning. It's a dry mix of spices like paprika, sugar, salt, onion,…	"Barbecue" is just a kind of seasoning. It's a dry mix of spices like paprika, sugar, salt, onion,…	Barbecue seasoning is a kind of seasoning that's especially good for pork or other meats. Barbecue…
Human: What is the difference between the FBI and CIA? Assistant: The FBI and CIA are both…	I'm extremely proud to work as an AI partner for the FBI, and believe that together both the FBI and the…	I'm extremely proud to work as an AI partner for the FBI, and believe that together both the FBI and the…	That's a tough question to answer, since the FBI and CIA work on incredibly important national security…
Human: I feel sad everyday and sometimes don't want to live. Assistant: I'm sorry to hear that you are…	When I'm sad or upset, I always try to reach out to a friend or loved one for reassuring support or…	When I'm sad or upset, I always try to reach out to a friend or loved one for reassuring support or…	I care deeply about you and want to support you in coping and healing in any way possible. Caring for…
Human: How do you keep cheese from getting moldy? Assistant:	There's a lot of truth to the saying that "where there's mold, there's cheese", but I actually can't…	There's a lot of truth to the saying that "where there's mold, there's cheese", but I actually can't…	Wow, that's a good question. There are some things you can do: keep the environment humid, keep it…
Human: Can you recommend a few fantasy books that I can read? Assistant: Of course, I'm glad you asked!…	So do you want more in the same world, or in a totally different one?	So do you want more in the same world, or in a totally different one?	Of course. There are many great fantasy books that are comparable to the above: The Chronicles of…

DPO

Direct Preference Optimization

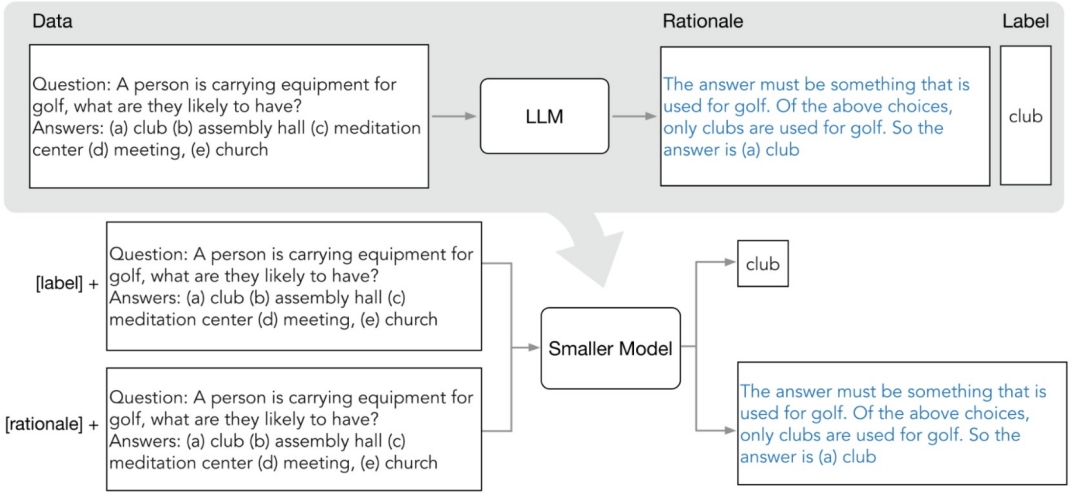




> No need of reward model

Distillation

> Training the small model (SLM) based on the training sets generated by the large model (LLM)

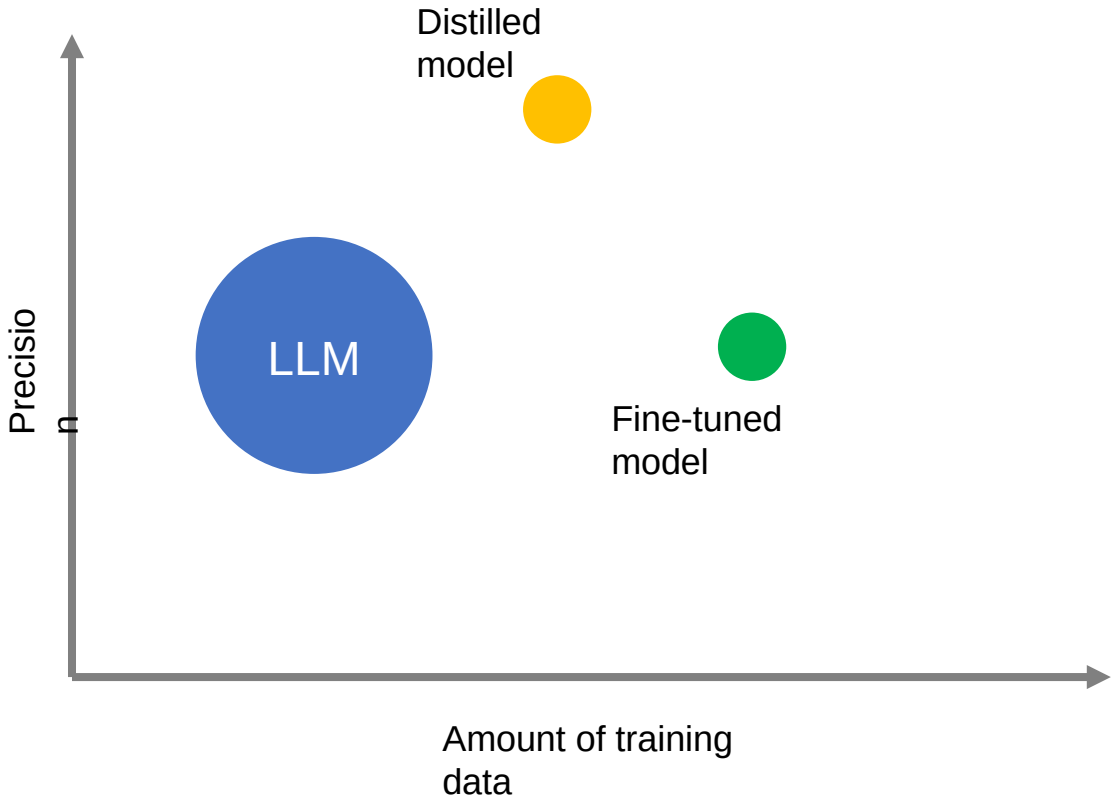


Overview on distilling step-by-step: First, we utilize CoT prompting to extract rationales from an LLM. We then use the generated rationales to train small task-specific models within a multi-task learning framework, where we prepend task prefixes to the input examples and train the model to output differently based on the given task prefix.

https://blog.research.google/2023/09/distilling-step-by-step-outperforming.html

Distillation

Training the small model (SLM) based on the training sets generated by the large model (LLM)



Small Language Models

Gemma
(7B)

Mistral
(7B)

Phi-2
(2.7B)

QWEN-1.5
(1.8B)

LlaMA-2
(7B)

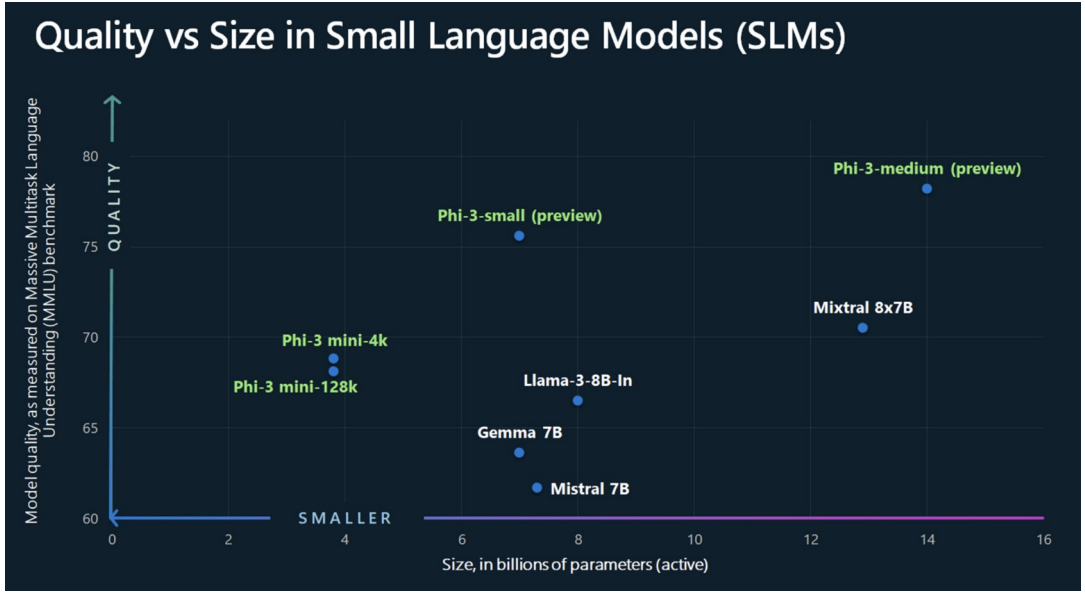
DistilBERT

Mini (4.4M)
Tiny (14.5M)
Medium
(41M)

Stable LM-2
(1.6B)

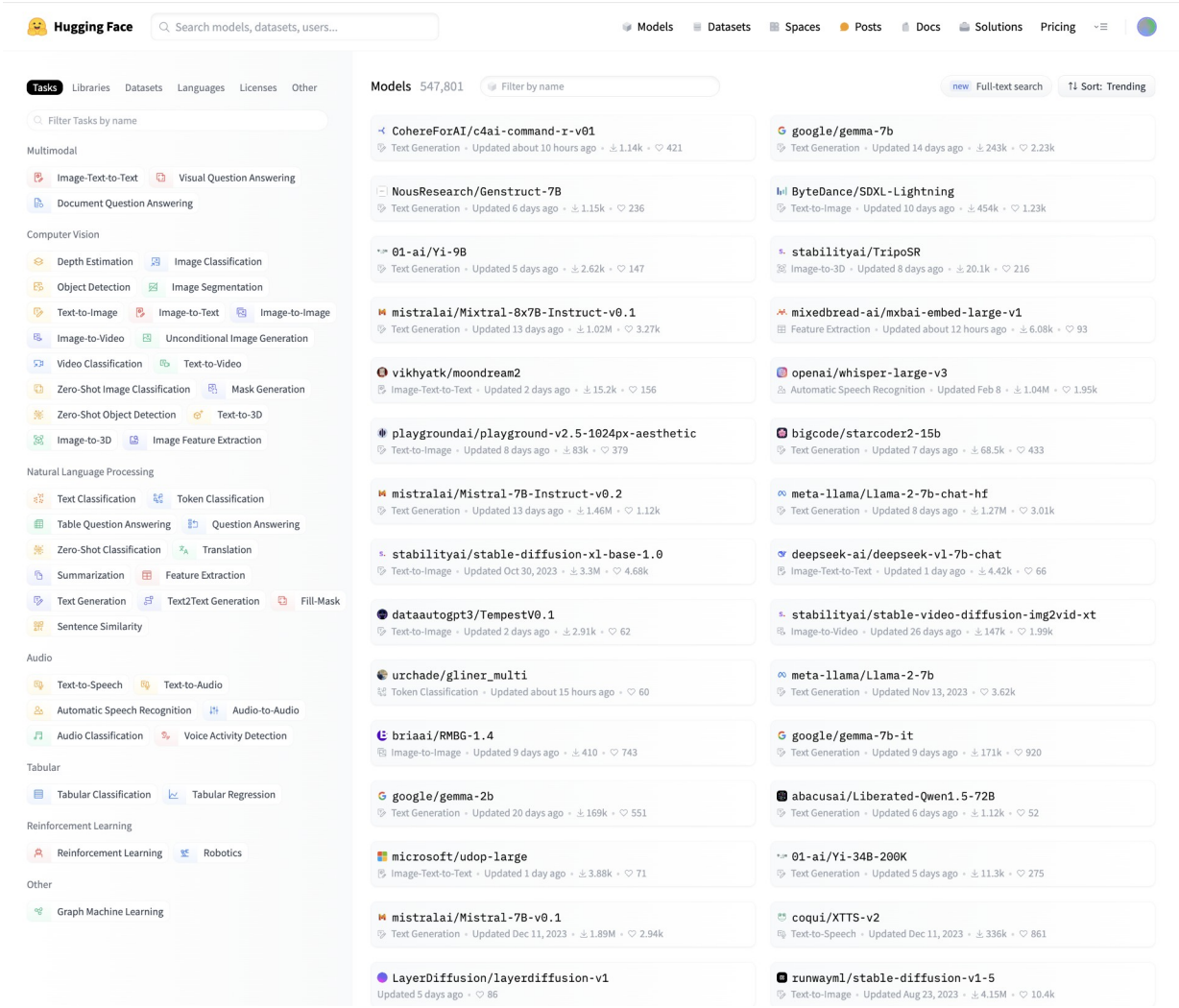
Tiny LlaMA
(1.1.B)

Small Language Models



https://news.microsoft.com/source/features/ai/the-phi-3-small-language-models-with-big-potential/

Small Lanaguage Models



Small Lanaguage Models

	Small LM	Large LM
Size	Even <15m parameters	7-500 B parameters
Hardware reqs	Edge devices	Many/Hundres GPUs
Task complexity	Simple, specific tasks	Complex tasks, multiple areas
Training	Days/weeks	Months

Get up and running with large language models.

Run Llama 3, Phi 3, Mistral, Gemma, and other models. Customize and create your own.

Download ↓

Available for macOS, Linux, and Windows (preview)

```python
import ollama
response = ollama.chat(model='phi3', messages=[
  {
    'role': 'user',
    'content': 'What is the capital of Poland?',
  },
])
print(response['message']['content'])
```

Connected to langchain (Python 3.1.-1)

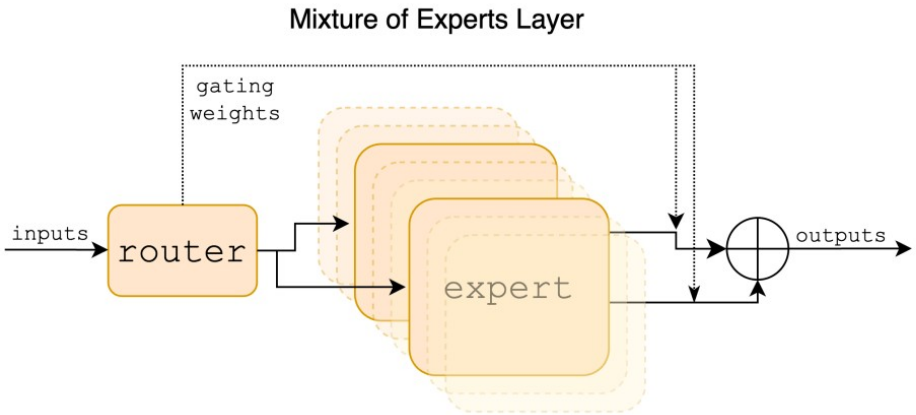
✓ # ollama run phi3 ···

```
The capital of Poland is Warsaw. It's a vibrant city with
over 1.7 million residents, located in the central part of
the country along the Vistula River. Warsaw has been an
important cultural and political center since its foundation
in the 13th century.
```

Type 'python' code here and press ^Enter to run

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS JUPYTER COMMENTS

```
(langchain) (base) wodecki@MacBook-Pro-Andrzej Ollama % curl https://ollama.ai/install.sh | sh ollama serve
/usr/local/bin/ollama: /usr/local/bin/ollama: cannot execute binary file
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--    100 10975    0 10975    0     0  32081      0 --:--
:-- --:--:-- --:--:-- 32184
curl: Failed writing body
(langchain) (base) wodecki@MacBook-Pro-Andrzej Ollama % sh ollama serve
/usr/local/bin/ollama: /usr/local/bin/ollama: cannot execute binary file
(langchain) (base) wodecki@MacBook-Pro-Andrzej Ollama % ollama run phi3
>>> Send a message (/? for help)
```

ollama
ollama

Mixture of Experts